Alcohols

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1 Alcohol Foundations

Core Concept 1.1: Alcohols

Alcohols (ROH) are a fundamental class of compounds characterized by the presence of one or more hydroxyl (–OH) groups attached to a carbon atom.

Because of this hydroxyl group(s) they possess relatively **high melting and boiling points** and **high water solubility** because of their ability to hydrogen bond. However, their water solubility decreases with the size of R because **R is hydrophobic** and **-OH is hydrophilic**.

The acidic nature of alcohols is represented by the shell equation $ROH + H_2O \rightleftharpoons RO^- + H_3O^+$. Using this equation, here are the pK_as of alcohols across various R groups. *As a reminder, lower pK_a values means higher acidity.*

#	ROH	pK_a
1	H_2O	15.7
2	CH_3OH	15.5
3	CH_3CH_2OH	15.9
4	$(CH_3)_2CHOH$	17.1
5	$(CH_3)_3COH$	18
6	$ClCH_2CH_2OH$	14.3
7	CF_3CH_2OH	12.4
8	$CF_3CH_2CH_2OH$	14.6

Table 1: pK_a of various alcohols

Core Concept 1.2: Solvation and Alcohol Acidity

Notice that as the alcohol shifts from methyl → primary → secondary → tertiary R groups (2 → 3 → 4 → 5), the acidity of the alcohol increases.

This pattern is primarily driven by **solvation** where the negative charge on an alkoxide ion (RO^-) is stabilized through solvent molecules surrounding the oxygen to spread out the charge. In more sterically hindered instances like $(CH_3)_3COH$, this solvation occurs to a lesser extent, making the alkoxide form less stable in comparison to methoxide (CH_3OH).

Core Concept 1.3: Inductive Effect and Alcohol Acidity

The larger presence of electronegative halogen species in ROH increases the acidity of the alcohol due to the **inductive effect**, which refers to the transmission of a partial electrical charge throughout a molecule's atoms. A resulting alkoxide RO^- is more stabilized when the R group is able to "share the burden" of the concentrated negative charge on the oxygen.

As a result, compound 7 (CF_3CH_2OH) has a smaller pK_a value compared to compound 6 ($ClCH_2CH_2OH$). This acidic pattern also occurs with increasing distance between the –O– and halogen species, an observation we see in the pK_a differences between compound 7 and 8.

2 Alcohol Synthesis

Alcohol is synthesized primarily through one of three different methods. We will go through each method in turn.

Core Concept 2.1: Radical Halogenation and S_N2 / S_N1 →Alcohols

Our goal can be summarized by the following chemical equation: $R-L + OH^- \longrightarrow R-OH + L^-$.

In the general absence of steric hinderance around the electrophile — such as H_3C-L or $R_{prim}-L$ — we can accomplish this reaction purely via S_N2. However, the possibility of undesired E2 reactions increases as steric hinderance becomes more present — such as in β branched compounds.

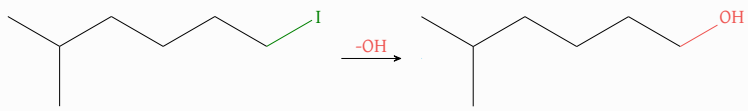

Figure 1: Alcohol synthesis via S_N2

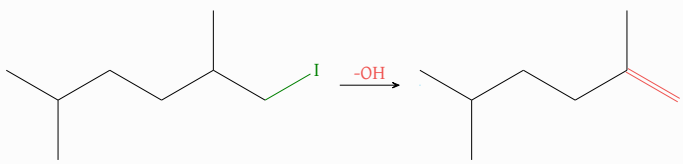

Figure 2: Undesired E2 reaction in β branched R groups

In order to resolve this E2 issue, we use a less basic, "masked" OH-equivalent such as acetate.

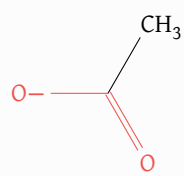

Figure 3: Acetate ion structure

After substituting the halogen leaving group in a β branched R group, we can convert the acetate substituent into a -OH group via hydrolysis.

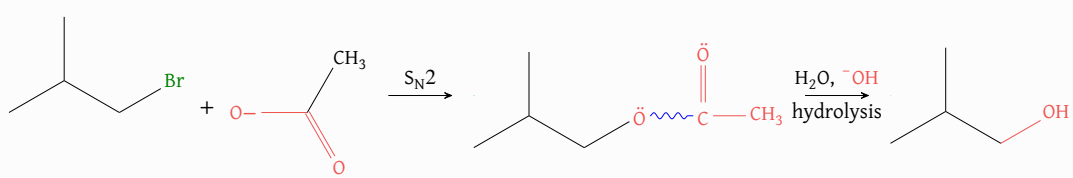

Figure 4: Alcohol synthesis via acetate substitution and hydrolysis

For $R_{sec}-L$ and $R_{ter}-L$, S_N2 is inefficient or impossible due to steric hinderance around the electrophile, thus we carry out alcohol synthesis via S_N1. By using a poor Nu like H_2O, we can avoid E2, however, we are again met with the possibility of E1 products. To reduce this undesired occurrence, we can lower temperature to minimize the free energy push the E1 reaction gets from having a greater entropy change (ΔS) than the S_N1 pathway.

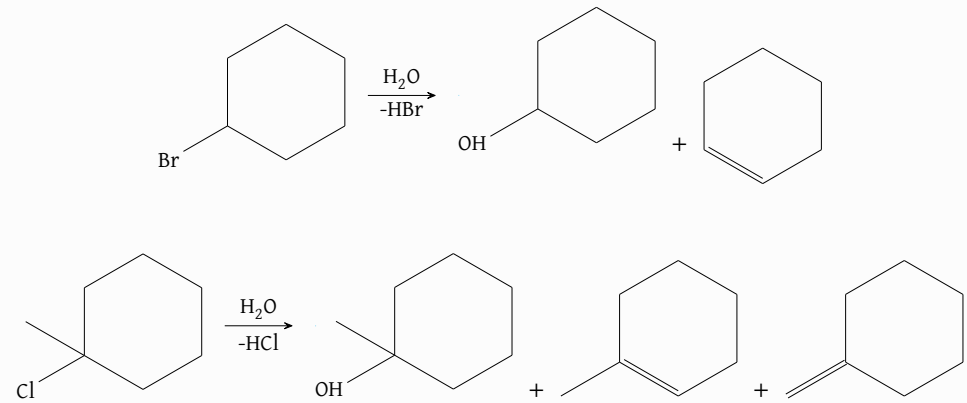

Figure 5: Alcohol synthesis via S_N1 with Br and Cl; undesired E1 products to the right can be minimized with lowering T

Core Concept 2.2: Reduction of Aldehydes and Ketones →Alcohols

With aldehydes or ketones, we must find a way to incorporate H_2 into the compound to produce a $R_{sec}-OH$ or $R_{prim}-OH$ alcohol, respectively.

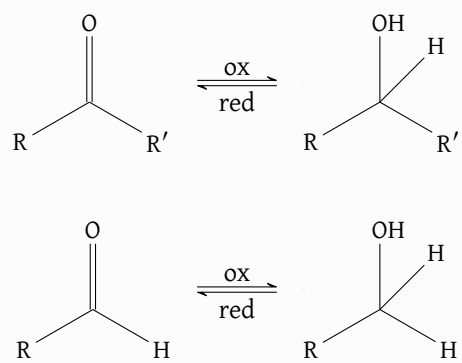

Figure 6: Reduction of aldehydes (*bottom*) and ketones (*top*) to alcohols

Direct hydrogenation is possible, but cumbersome because it needs H_2 pressure and catalysts (e.g. Rd, Pt). Instead, the most common laboratory method is via **hydride** (H^-) donors, which are prepared either through boron or aluminum via the following reactions.

$Na^+H^- + BH_3 \longrightarrow Na^+BH_4{}^-$ *sodium borohydride*

$Li^+H^- + AlH_3 \longrightarrow Li^+AlH_4{}^-$ *lithium aluminum hydride*

$BH_4{}^-$ and $AlH_4{}^-$ are still basic by nature and will undergo protonation by alcohols or water (protic) to give H_2, an undesired reaction pathway. However their reactivities differ. $AlH_4{}^-$ is a much more aggressive H^- donor compared to $BH_4{}^-$ because Al is less electronegative than B, thus the nearby negative charge from H^- is less stable compared to B. Consequently, $AlH_4{}^-$ is so reactive that it will react even with air moisture to form H_2.

As a result, $Na^+BH_4{}^-$ **allows us to use alcohol solvents** if needed without worrying about protonation. On the other hand, $Li^+AlH_4{}^-$ requires aprotic or dry solvents to prevent protonation, such as ethers.

$Na^+BH_4{}^- \xrightarrow{H_2O} 4\,H_2 + Na^+B(OH)_4{}^-$ *slow*

$Li^+AlH_4{}^- \xrightarrow{H_2O} 4\,H_2 + Li^+Al(OH)_4{}^-$ *violent*

The donated H^- will thus attack the central carbon of the ketone or aldehyde, resulting in an alkoxide, which is then protonated by an alcohol solvent (for $Na^+BH_4{}^-$) or aqueous work up (for $Li^+AlH_4{}^-$). *Mind that the aqueous work up is in a separate step after purification. test change*

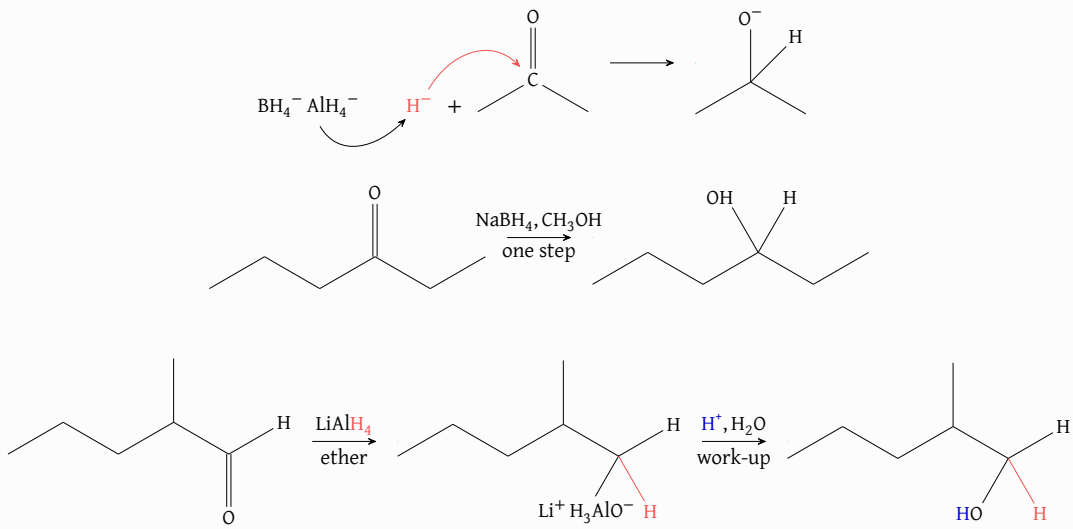

Figure 7: Formation of alkoxide intermediate (*top*); reduction of ketones and aldehydes via $NaBH_4$ (*middle*) and $LiAlH_4$ (*bottom*)

In all, the mechanism of $NaBH_4$ is termolecular, concerted, and simultaneous. On the other hand, $LiAlH_4$ is a stepwise delivery of [1] H^- and [2] H^+, H_2O (the work-up step).

In order to **oxidize the resulting alcohols back to aldehydes and ketones**, we can use Cr^{VI} reagents, which accept e^-s and consequently form Cr^{III}. One such compound is chromic acid, formed via the following.

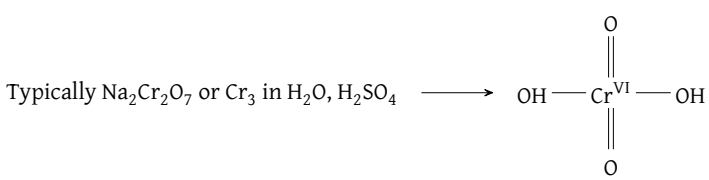

Figure 8: Formation of chromic acid reagent

Chromic acid is best for oxidizing $R_{sec}-L$, which results in ketones. $R_{prim}-L$, on the other hand, are **overoxidized** in aqueous medium.

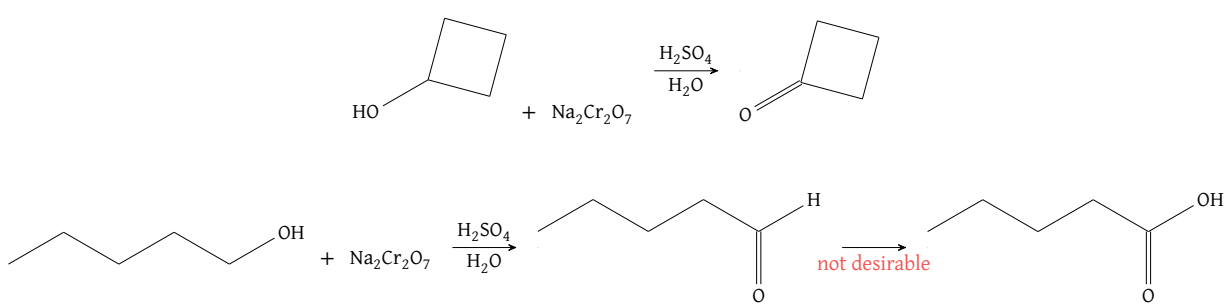

Figure 9: Oxidation of alcohols via chromic acid

Thus to oxidize $R_{prim}-L$, we instead use pyridinium chlorochromate (PCC) which is soluble in organic solvents. This compound avoids the overoxidation that chromic acid would have performed.

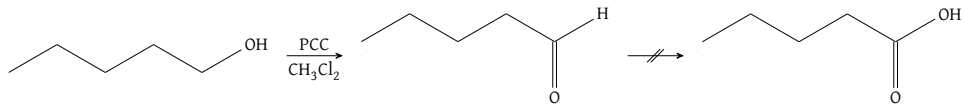

Figure 10: Pyridinium Chlorochromate (PCC) reaction

Core Concept 2.3: Organometallics →Alcohols

Alcohol can be synthesized from **organometallics**, namely **grignard reagents** (RMgX) and organolithium reagents (RLi). Just as BH_4^- and LiH_4^- are were used to donate H^- to push the π bond e^-s onto the oxygen, grignard reagents do the same but donates an entire R^- rather than H^-.

We can synthesize organolithium reagents and grignard reagents via the following chemical pathways.

$$\text{Li} + \text{RX} \xrightarrow[-\text{X}]{\text{hexane}} \text{R}-\text{Li} + \text{LiX} \textit{ organolithium reagents}$$

$$\text{Mg} + \text{RX} \xrightarrow{\text{ether}} \text{R}-\text{Mg}-\text{X} \textit{ grignard reagents}$$

Due to electronegativity differences organolithium reagents must be done in a nonreactive, nonpolar solvent like hexane. Grignard reagents, on the other hand, must be done in ethers to prevent RMgX from clumping up via solvation.

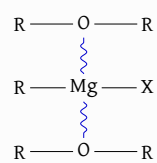

Figure 11: Solvation of grignard reagent in ether

The resulting R – M(X) bond is highly polarized due to the EN difference between the organic R group and metal. Unlike most organic molecules — where carbon is bonded to more EN elements and bears a partial positive charge — the carbon in R – M(X) is more EN than its metallic partner. This reversal of typical polarity, or **umpolung**, renders the R group carbon both strongly basic and nucleophilic, making them extremely easy to protonate via hydrolysis.

$$\text{R}^{\delta+} - \text{X}^{\delta-} \longrightarrow \text{R}^{\delta-} - \text{M}^{\delta+}(\text{X})$$

Figure 12: Polarization of haloalkanes (*left*) versus grignard reagents (*right*)

Consequently, they must be made in very dry solvents unless the goal was to convert a R – X to a R – H.

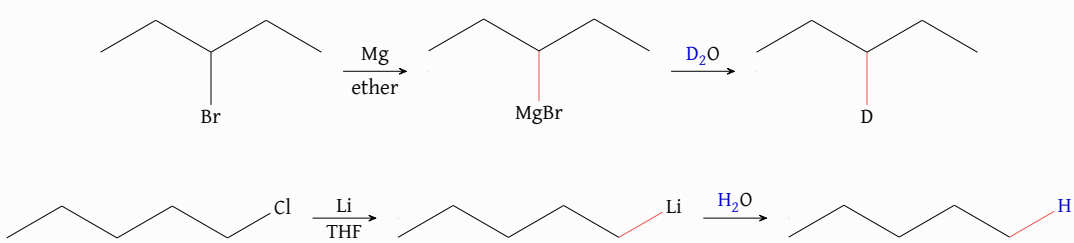

Figure 13: Conversion of haloalkanes to alkanes via organometallics

However, when reacted with ketones, aldehydes, or formaldehyde, they form cooresponding alkoxides, which can then be put through an aqueous work-up in order to form alcohol. This method of alcohol formation allows chemists to add a R group to any carbonyl containing compound whereas the previously covered methods could only add H.

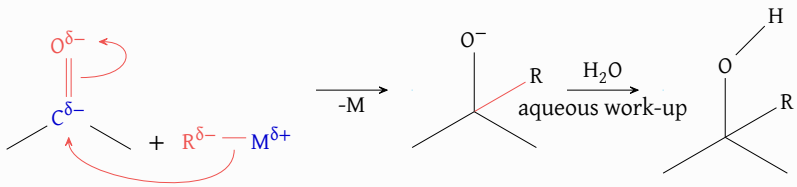

Figure 14: Alcohol synthesis via grignard addition to carbonyls

3 Alcohol Reactions

Deprotonating alcohol requires a stronger base than RO^-. In other words, the base must have a stronger affinity to the H^+ than the alkoxide (RO^-). If the alcohol is acidic, water will usually suffice in this matter. However, in cases where water is more acidic than the alcohol itself, H^+ can be donated to the alcohol and a subsequent H_2O L group will be formed, leaving the resulting product to have a highly reactive, empty atomic orbital on the relevant C after the H_2O L group departs.

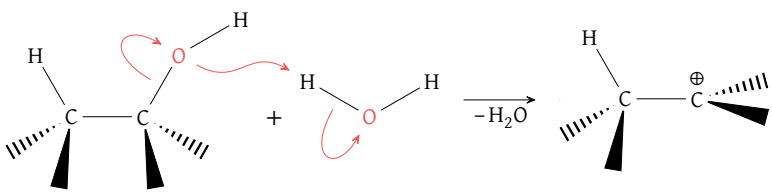

Figure 15: Formation of oxonium ion and carbocation

This phenomenon presents a range of complications when dealing with alcohol reactions. For instance, let's say that we want to replace the $-OH$ with a different functional group like a halide X. This can occur either through S_N2 or S_N1 .

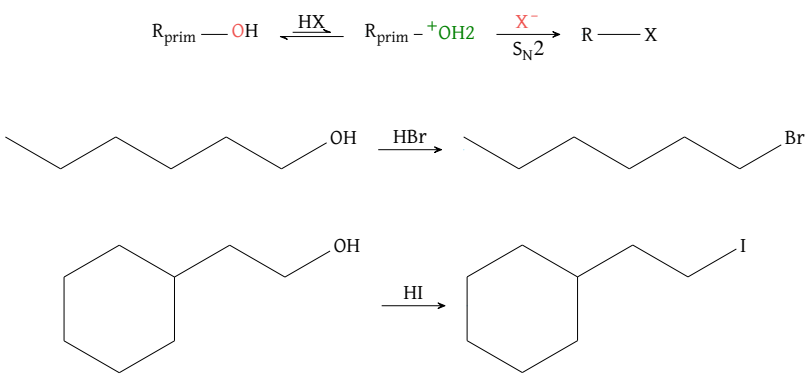

Figure 16: S_N2 of alcohols; needs strong H^+ with a good nucleophile HCl, HBr, HI

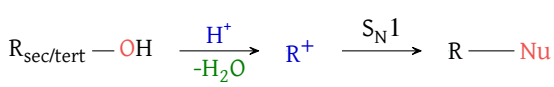

Figure 17: S_N1 of alcohols; *E1 may occur as well, leading to an alkane end product, but* S_N1 *dominates for* R_{sec} *in* H_2O

However, mixtures ensue due to several different mechanisms.

Core Concept 3.1: Hydride Shifts

Hydride shifts refer to the movement of a intramolecular hydrogen towards a neighboring empty p orbital.

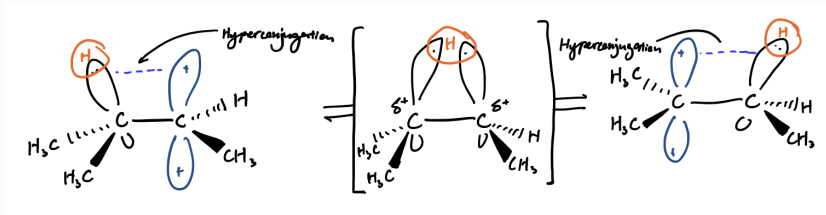

Figure 18: Hydride Shift Mechanism in SN1 Reaction

This is particularly an issue if the shift is favorable, such as the move from a R_{sec}^+ to a R_{tert}^+. Degenerate shifts ($R_{sec}^+ \rightarrow R_{sec}^+$ or $R_{tert}^+ \rightarrow R_{tert}^+$) are less common, but can still occur, resulting in the halide attacking at a unintended location.

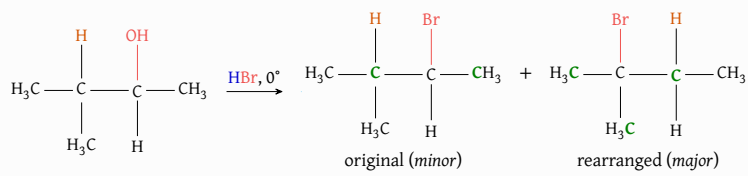

Figure 19: Hydride shift resulting in regioisomeric products; the original structure (*left*) would be a minor product whereas the rearranged structure (*right*) would be the major product

In specific cases like ring structures, these hydride shifts happen stereospecifically. The bottom line is that **mixtures are to be expected.**

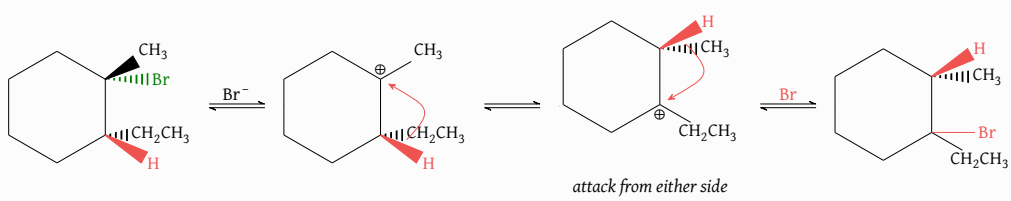

Figure 20: Stereospecific hydride shift in ring structures

Core Concept 3.2: E1 Reactions

E1 reactions become particularly more prevalent at higher temperatures and in the absence of a viable nucleophile. *Notice how the H^+ detachment at the last step happens at the tertiary position and not the secondary due to greater stablity from more hyperconjugation.*

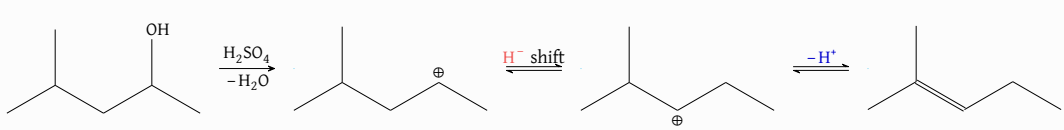

Figure 21: E1 mechanism in alcohols

Core Concept 3.3: Alkyl Shifts

Additionally, entire alkyl or R shifts can occur, although not as commonly as hydride shifts because of the greater E_a. These shift can occur even in $R_{prim}-OHs$.

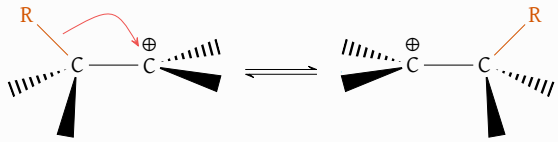

Figure 22: Alkyl shift mechanism

However, alkyl shifts are especially favorable and fast when relieving ring stain.

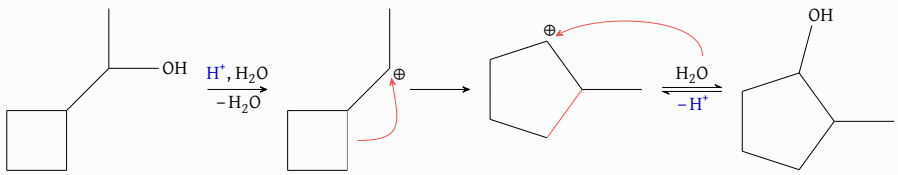

Figure 23: Ring strain relieving alkyl shift

So, how do we actually turn an ROH into RNu without all of these potential issues that are bound to happen? The solution is to use inorganic derivatives as reactive or isolable intermediates.

Core Concept 3.4: Substituting OH with Br

Chemists use PBr_3 to replace OH groups with Br: $3\,ROH + PBr_3 \longrightarrow 3\,RBr + H_3PO_3$. The mechanism for this reaction is as follows.

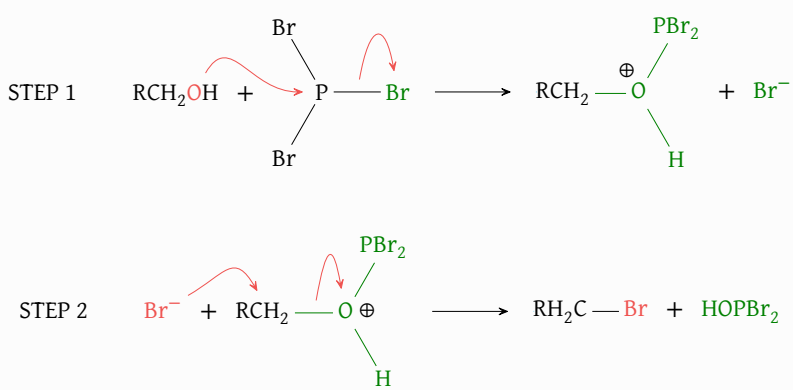

Figure 24: Mechanism of alcohol bromination using PBr_3; *STEP 1 replaces the bad leavin group OH with a good leaving group in order to prepare for the* S_N2 *in STEP 2*

Core Concept 3.5: Substituting OH with Cl

Chemists use $SOCl_2$ to replace OH groups with Cl: $ROH + SOCl_2 \longrightarrow RCl + SO_2 + HCl$. The mechanism for this reaction is as follows.

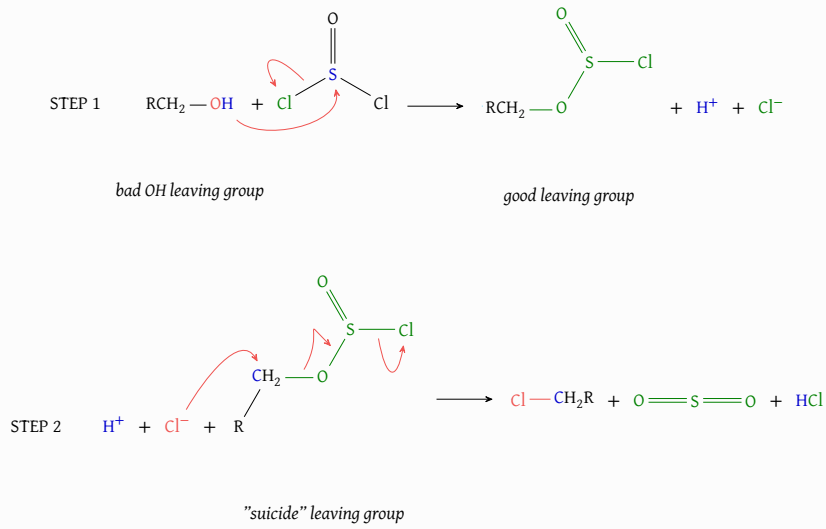

Figure 25: Mechanism of alcohol chlorination using $SOCl_2$

It is important to note here that pyradine is used as a solvent in this reaction in order to neutralize the HCl byproduct from further protonating the alcohol or the product, effectively preventing unintended compounds from forming.

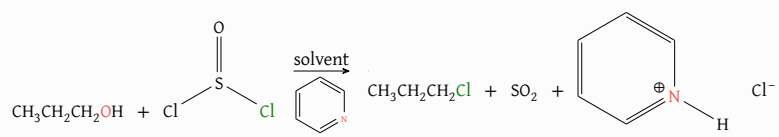

Figure 26: Reaction of alcohol with $SOCl_2$

Core Concept 3.6: Substituting OH with Good Leaving Groups

To substitute OH with a good leaving group, chemists use isolable sulfonates such as the following molecules and cooresponding reactions.

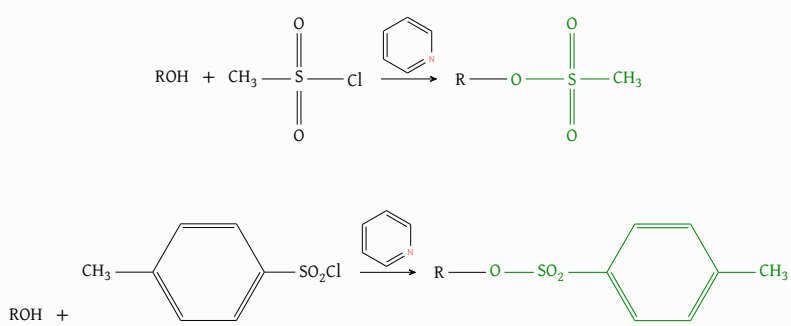

Figure 27: Formation of isolable sulfonates

Notice that the displaced Cl^- after the sulfonates attach to the R group is stabilized as a floating ion via solvation by the pyradine solvent as seen in the reaction previously shown with $SOCl_2$ in *Core Concept 3.5*.